

Draw the tangent ZPR at P — the straight path the body WOULD take if the force suddenly vanished; this line is the heart of the argument. The distance by which the curve pulls away from that tangent is the versed sine QR , drawn from the nearby point Q back toward the tangent. By Cor. 4 of Prop. I this fall in a given time is directly as the force. And by Cor. 2 and 3 of Lem. II, lengthening the time in any ratio augments the arc in that ratio and the versed sine in the DUPLICATE ratio — so the versed sine is as the force times the square of the time.